

A SHIFT IN SOFTWARE ABSTRACTION

High-level languages

Fortran, C

OS abstractions

processes, virtual memory

ML workloads

deep learning

Foundation models

Software 3.0: prompt a model

each shift pulls an architecture response

THE ARCHITECTURE RESPONSE

RISC instruction sets

with optimizing compilers

Processor features

protection, privilege levels

Specialized accelerators

TPUs, NPUs

Architecture 2.0

the design loop itself

changes WHAT architects build

changes HOW architects build

Prior waves changed the artifact; foundation models change the design loop.